4 In a certain country it is known that 11% of people are left-handed.

(a) Calculate the probability that, in a random sample of 98 people from this country, 5 or fewer are found to be left-handed, giving your answer correct to 3 significant figures. **[1]**

An anthropologist believes that the proportion of left-handed people is lower in a particular ethnic group.

The anthropologist collects a random sample of 98 people from this particular ethnic group in order to test the hypothesis that the proportion of left-handed people is less than 11%.

The anthropologist carries out the test at the 1% level.

(b) Determine the critical region for this test. **[3]**

4(a)	
4(b)	